

MD ABDUR RAZZAQ RIYADH

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EDUCATION

University of the Basque Country

San Sebastian, Spain

Ph.D. Researcher

Sep 2025 – Present (expected 2029)

- Unsupervised ASR and TTS with speech language models: learning speech-text mapping without paired supervision, targeting languages where labelled audio is scarce.
- Single-node, multi-GPU training with PyTorch FSDP on SLURM clusters.
- Supervised by Eneko Agirre and Eva Navas

University of the Basque Country

San Sebastian, Spain

M.Sc. in Language and Communication Technologies

Sep 2024 – Jun 2025

- Erasmus Mundus Joint Master's (EMJMD), full scholarship; double degree with University of Malta.
- GPA 9.63/10.
- Coursework: Speech Technologies, NLP Applications, Theoretical Linguistics
- Thesis: SpeechLM for ASR in Low-Resource Languages (see Research Experience).

University of Malta

Msida, Malta

M.Sc. in Human Language Science and Technology

Sep 2023 – Jun 2024

- EMJMD in Language and Communication Technologies; year one of the double degree above.
- Coursework: AI Numerical Methods, Deep Learning Approaches to NLP, Statistics for Data Scientists

University of Asia Pacific

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering

Sep 2016 – Jun 2020

- CGPA 3.70 out of 4.00.

PUBLICATIONS

Riyadh, M. A. R., Agirre, E., Navas, E., & Borg, C. (2026). SpeechLM for Automatic Speech Recognition in Low-resource Languages. In Proceedings of Speech Language Models in Low-Resource Settings: Performance, Evaluation, and Bias Analysis (SPEAKABLE) @ LREC 2026, pp. 125–131. **Best Paper Award.** [ACL Anthology] [Code]

Riyadh, M. A. R., & Nabhani, S. (2024). Mela at AraIEval Shared Task: Propagandistic Techniques Detection in Arabic with a Multilingual Approach. In Proceedings of the Second Arabic Natural Language Processing Conference, pp. 478–482. [ACL Anthology] [Code]

Nabhani, S., & **Riyadh, M. A. R.** (2024). Mela at CheckThat! 2024: Transferring Persuasion Detection from English to Arabic — A Multilingual BERT Approach. CLEF (Working Notes), pp. 546–551. [CEUR-WS]

Abela, K., **Riyadh, M. A. R.**, Galea, M., Busuttill, A., Kovalev, R., Williams, A., & Borg, C. (2024). UOM-Constrained IWSLT 2024 Shared Task Submission — Maltese Speech Translation. Proceedings of the 21st International Conference on Spoken Language Translation (IWSLT 2024), pp. 108–113. [ACL Anthology]

RESEARCH EXPERIENCE

SpeechLM for ASR in Low-Resource Languages

San Sebastian, Spain & Msida, Malta

M.Sc. Thesis — HiTZ Zentroa / Univ. of Malta

Jan 2025 – Jun 2025

- Fine-tuned and evaluated multimodal LLaMA-Omni on Basque/Maltese ASR, then built “Latxa-Omni”, a Whisper encoder bridged to Latxa, a Basque-adapted LLM, by porting the LLaMA-Omni architecture to a new language backbone.
- Trained the speech adaptor and the 8B LLM on 676 h of Basque speech on 4 GPUs (single node; DeepSpeed ZeRO-2, bf16), with the Whisper encoder and speech decoder frozen throughout, cutting Basque WER from 10.43% to 8.09%; swapping in the Basque-adapted Latxa pulled it down further to 7.98%, the best result reported in the paper.
- Isolated data scale as the driver: on 35 h of Maltese the same recipe *lost* to the Whisper baseline (16.74% → 25.65% WER), and a 35 h Basque control reproduced the failure (10.43% → 18.37%) — the gain requires hundreds of hours, not the architecture.

UOM-Constrained IWSLT 2024 Shared Task: Maltese Speech Translation

Msida, Malta

Shared Task Submission

Feb 2024 – Apr 2024

- Built a cascaded ASR → MT pipeline for the constrained track, where no pre-trained models were permitted: trained a DeepSpeech-v1 architecture (64-unit hidden layers vs. the default 2048), a KenLM 3-gram rescorer and a BiLSTM seq2seq from scratch on a 12-hour speech budget.
- Selected the BiLSTM over Transformer-base/large on dev BLEU (25.76 vs. 24.54 / 25.20) and shipped it as the official submission; 12 h of speech capped ASR at 97.15% WER (HMM baseline 112.33%), bounding end-to-end translation at 0.5 BLEU.

AraIEval & CheckThat! 2024: Arabic Propaganda & Persuasion Detection

Msida, Malta

Shared Task Submission

Apr 2024 – May 2024

- Fine-tuned mBERT as a multi-label token tagger for Arabic propaganda-technique detection, placing 2nd of 6 teams with a 10th-layer classifier head at 0.28 micro-F1 against a 0.01 shared-task baseline; a layer ablation showed 8th-layer features beat final-layer ones by 0.09 micro-F1 (0.33 vs. 0.24, +38% relative). Adapted the setup for English-to-Arabic persuasion detection.

EXPERIENCE

HiTZ Zentroa (UPV/EHU)	San Sebastian, Spain
<i>Research Intern</i>	<i>Nov 2024 – Jan 2025</i>
Explored and evaluated candidate SpeechLM architectures for Basque ASR, selecting the whisper-large-v3 + LLaMA 3.1-8B-Instruct setup later used in the thesis.	
Brac IT	Dhaka, Bangladesh
<i>Software Engineer</i>	<i>May 2023 – Jun 2023</i>
- Engaged in the development and ongoing maintenance of a financial platform which enables BRAC to run a prominent micro-credit program in Bangladesh. - Stack: Java, Spring Framework & Spring Boot, PostgreSQL, Docker.	
CMED Health Ltd	Dhaka, Bangladesh
<i>Software Engineer</i>	<i>Oct 2020 – Apr 2023</i>
- Designed and developed robust RESTful APIs in Java to support mobile and web clients. - Stack: Java, Spring Framework & Spring Boot, Seam Framework, Apache Tomcat, JBOSS AS, MariaDB, MySQL.	

TECHNICAL SKILLS

Machine Learning & Speech/NLP: PyTorch, Hugging Face Transformers, DeepSpeed (ZeRO), PyTorch FSDP, mixed-precision (bf16) training, Whisper, LLaMA
Training Infrastructure: SLURM, single-node / multi-GPU training, Docker, Git, Linux
Programming: Python, Java, SQL

AWARDS

2026 – Best Paper Award, SPEAKABLE Workshop @ LREC 2026.
2023 – Erasmus Mundus Joint Master’s Degree (EMJMD) full scholarship, Language and Communication Technologies (LCT) — competitive, EU-funded.
2018, 2019 – Dean’s Honor List, University of Asia Pacific